Two Reparameterizations of the Region Successor Measure: Temporal Differencing and a Fixed-Partition Simplex

Working note — Factored-FB project
2026-08-06, v1 (derivations + first cube-single wave)

Abstract

Learning the region successor measure by temporal-difference regression on the occupancy indicator fails on manipulation play data in a specific way: the targets are dominated by an action-independent dwell term and the critic converges to the constant that term implies. This note states two reparameterizations that leave the object and its greedy policy unchanged while changing what is regressed. Delta regresses the temporal difference of the indicator; we prove the identity $J = (1-\gamma)M - h(x_0)$, show it preserves the action $\arg\max$, and show that the estimator's degenerate constant relocates to zero rather than disappearing. Part replaces the scalar measure by a distribution over a fixed K-cell partition fitted by cross-entropy, making validity structural and supervision dense. On cube-single-play-v0, Delta exhibits precisely the zero-flux collapse the analysis predicts, while Part reaches 0.884 goal success and transfers zero-shot to point-goal reward families. Both are alternatives to the multi-step operator of the companion note: that one keeps the target and lengthens the horizon; these keep the horizon and change the target.

1 Setting and the object

Offline unlabeled "play" data $\mathcal{D}$ from OGBench (Park et al., 2025), state observations. Write $x = \phi_{\text{obj}}(s)$ for the object coordinates carried by the observation (for cube-single, the cube position $s_{19:22}$). For a goal g and tolerance ε define the region indicator

$$h_g(x) \;=\; \mathbf{1}[\|x - g\| \le \varepsilon] \;\in\; \{0, 1\}, \tag{1}$$

extended to a conjunction of blocks with per-block tolerances for multi-object environments. Under the goal-conditioned policy $\pi(\cdot \mid \cdot, g)$ the object of interest is the ball-visitation region successor measure (Dayan, 1993; Blier et al., 2021; Touati & Ollivier, 2021)

$$M(s, a, g) \;=\; \mathbb{E}\Big[\textstyle\sum_{t\ge 0} \gamma^t\, h_g(x_{t+1}) \;\Big|\; s_0 = s,\; a_0 = a\Big], \tag{2}$$

the expected discounted time spent inside the region, with rewards counted at the next state. Its Bellman equation is

$$M(s, a, g) \;=\; \mathbb{E}\big[\, h_g(x_1) \;+\; \gamma\, M(s_1, a_1, g)\,\big], \qquad a_1 \sim \pi(\cdot \mid s_1, g), \tag{3}$$

and the estimator regresses $\delta = M(s, a, g) - h_g(x') - \gamma\, \mathrm{sg}[\bar{M}(s', a', g)]$ against a batch of columns. Note $M \in [0, \frac{1}{1-\gamma}]$, which is $[0, 100]$ at $\gamma = 0.99$.

The defect. A trajectory that enters the region at step T and leaves at $T + L$ contributes L identical labels $h_g = 1$ to equation 3. The mass of the target is thus governed by the dwell length L, a property of the region and the data rather than of the action taken at (s, a). The fitted critic settles near the constant $c^\star$ solving $c = \mathbb{E}[h] + \gamma c$,

$$c^\star \;=\; \frac{\mathbb{E}_{\mathcal{D}}[h]}{1 - \gamma}, \tag{4}$$

which is large, well scaled, and action-independent, and from which the actor extracts nothing. The companion note attacks this by lengthening the horizon of the operator so the dwell mass enters the observed term. The two variants below instead keep the one-step operator and change what is predicted. Neither changes M.

2 Variant A (Delta): temporal differencing of the indicator

2.1 Definition and the exact identity

Replace the target reward in equation 2 by the temporal difference of the indicator,

$$J(s,a,g) \;=\; \mathbb{E}\Big[\textstyle\sum_{t\geq 0}\gamma^t\big(h_g(x_{t+1})-h_g(x_t)\big)\Big]. \tag{5}$$

Proposition 1 (Reparameterization). For every (s,a,g) and every policy, with both series counted under the convention of equation 2,

$$J(s,a,g) \;=\; (1-\gamma)\,M(s,a,g) \;-\; h_g(x_0), \qquad x_0=\phi_{\mathrm{obj}}(s). \tag{6}$$

Proof. Split equation 5. The first series is M by definition. For the second, re-index and use that x_0 is deterministic given s:

$$\sum_{t\geq 0}\gamma^t h_g(x_t) = h_g(x_0)+\sum_{t\geq 1}\gamma^t h_g(x_t) = h_g(x_0)+\gamma\sum_{t\geq 0}\gamma^t h_g(x_{t+1}).$$

Taking expectations the second series equals $h_g(x_0)+\gamma M$, so $J = M-\big(h_g(x_0)+\gamma M\big) = (1-\gamma)M - h_g(x_0)$. □

Corollary 1 (Policy invariance). $h_g(x_0)$ depends on (s,g) but not on a, and $1-\gamma>0$; hence $\arg\max_a J(s,a,g) = \arg\max_a M(s,a,g)$ for every (s,g). J is a reparameterization of the same object, not a different objective.

Corollary 2 (Boundedness). Since $M\in[0,\frac{1}{1-\gamma}]$ and $h_g(x_0)\in\{0,1\}$, we have $(1-\gamma)M\in[0,1]$ and $J\in[-1,1]$, uniformly in γ.

2.2 Estimator

J obeys the Bellman equation obtained from equation 3 by substituting the signed reward,

$$J(s,a,g) \;=\; \mathbb{E}\big[\big(h_g(x_1)-h_g(x_0)\big) \;+\; \gamma\, J(s_1,a_1,g)\big], \tag{7}$$

so the estimator differs from the baseline in exactly one term:

$$\underbrace{\delta = M - h_g(x') - \gamma\,\mathrm{sg}[\bar{M}]}_{\text{visitation}} \quad\longrightarrow\quad \underbrace{\delta = M - \big(h_g(x')-h_g(x)\big) - \gamma\,\mathrm{sg}[\bar{M}]}_{\text{delta}}. \tag{8}$$

The per-transition target takes four values: 0 for outside → outside, +1 on entry, 0 throughout a dwell, and −1 on exit. The counting convention is load-bearing — $h_g(x')$ on next_observations and $h_g(x)$ on observations. An off-by-one there silently learns a different object while leaving every loss curve plausible, and is pinned by a regression test.

2.3 What the differencing does to the supervision

Lemma 1 (Dwell telescoping). For a trajectory entering the region at step T and leaving at $T+L$, the contributions to the two targets are

$$\underbrace{\gamma^T\,\frac{1-\gamma^L}{1-\gamma}}_{\text{visitation: } L \text{ labels}} \qquad \text{and} \qquad \underbrace{\gamma^T(+1)+\gamma^{T+L}(-1) = \gamma^T\big(1-\gamma^L\big)}_{\text{delta: 2 labels}},$$

which agree exactly after the factor $(1-\gamma)$ of Proposition 1.

The dwell length is therefore not discarded but re-encoded as the discounted gap between an entry and an exit, rather than L repeated identical labels. The action-independent bulk leaves the regression while the information it carried is retained.

2.4 Relocation, not elimination, of the degenerate solution

Proposition 2 (Zero-flux constant). The constant critic consistent with equation 7 is

$$c_J^\star \;=\; \frac{\mathbb{E}_{\mathcal{D}}[\Delta h]}{1-\gamma}, \qquad \Delta h = h_g(x') - h_g(x). \tag{9}$$

Over a dataset in which every entry into a region is eventually matched by an exit, $\mathbb{E}_{\mathcal{D}}[\Delta h] \approx 0$, hence $c_J^\star \approx 0$.

Comparing equation 4 with equation 9, the lazy solution does not disappear: it moves. Its diagnostic value changes qualitatively, though. The visitation constant is large and well scaled and can be mistaken for a calibrated measure; the delta constant is ≈ 0, so a critic that has learned nothing announces itself. We pre-registered msq $= \mathbb{E}[M^2] \to 0$ with flat loss and behavior-cloning-floor success as the zero-flux collapse signature.

3 Variant B (Part): fixed-partition simplex measure

3.1 Definition

Fix K prototypes $\{p_k\}_{k=1}^K$ drawn from the dataset's object coordinates and define a soft partition by a per-block scaled radial-basis softmax,

$$\varphi_k(x) \;=\; \frac{\exp\!\big(-\tfrac{1}{2}\sum_b \|(x^{(b)} - p_k^{(b)})/\varepsilon_b\|^2\big)}{\sum_{k'} \exp\!\big(-\tfrac{1}{2}\sum_b \|(x^{(b)} - p_{k'}^{(b)})/\varepsilon_b\|^2\big)} \;\in\; \Delta^K, \tag{10}$$

where b indexes the blocks of the hit specification and ε_b is that block's tolerance, so heterogeneous tolerances contribute comparably. The learned object is the normalized occupancy over cells,

$$p^\star(s,a,g) \;=\; (1-\gamma)\,\mathbb{E}\Big[\textstyle\sum_{t\ge 0}\gamma^t \varphi(x_{t+1})\Big] \;\in\; \Delta^K. \tag{11}$$

The normalization is exactly the factor of Proposition 1: the k-th coordinate of equation 11 is $(1-\gamma)M$ integrated against cell k. Since $\sum_k \varphi_k \equiv 1$, total mass is conserved by construction and $p^\star$ lies on the simplex with no penalty enforcing it.

3.2 Estimator

A network emits K logits from (s,a,g) with $p = \mathrm{softmax}$; the target mixes the observed cell with the target network's prediction at a bootstrap action $a^+ \sim \pi(\cdot \mid s', g)$,

$$p_{\mathrm{tgt}} = \mathrm{sg}\big[(1-\gamma)\,\varphi(x') \;+\; \gamma\,\bar{p}(s', a^+, g)\big] \;\in\; \Delta^K, \tag{12}$$

$$\mathcal{L} = -\,\mathbb{E}\Big[\textstyle\sum_k p_{\mathrm{tgt},k}\,\log p_k\Big]. \tag{13}$$

Both terms of equation 12 lie on Δ^K and the weights sum to one, so $p_{\mathrm{tgt}} \in \Delta^K$ and equation 13 is a proper cross-entropy between distributions. Two properties follow that the scalar estimator lacks: validity ($p \ge 0$, $\sum_k p_k = 1$) is structural rather than penalized, and every transition supplies a dense K-vector target instead of a mostly-zero scalar.

3.3 Control and zero-shot reward evaluation

The actor maximizes the mass its own goal's cell receives,

$$q(s,a,g) \;=\; p(s,a,g)^\top z_g, \qquad z_g = \mathrm{sg}\big[\varphi(g)\big], \tag{14}$$

evaluated at frozen head parameters so that no actor gradient reaches the measure. By equation 11, $q = (1-\gamma)\,M \cdot z_g$ up to the action-independent level $\varphi(x_0)^\top z_g$, whence a centered normalizer is required.

For an arbitrary reward r presented at test time, evaluate it at the prototypes and read the value off with no retraining:

$$z_{r,k} = r(p_k), \qquad F^\Delta = p - \varphi(x_0), \qquad Q_r(s,a) \;=\; \frac{r(x_0) + F^{\Delta\top} z_r}{1-\gamma}. \tag{15}$$

Equation equation 15 is the variant-B analogue of equation 6: the $(1-\gamma)^{-1}$ and the $-\varphi(x_0)$ are the same normalization and the same action-independent offset, in vector form.

Conversion for variant A. A checkpoint trained under equation 8 represents J, not M; any calibration or reward readout must invert equation 6 first,

$$\widehat{M} \;=\; \frac{h_g(x_0) + \widehat{J}}{1-\gamma}. \tag{16}$$

For a composite task vector $z_r = \sum_j w_j \psi(g_j)$ the correct offset is the same-weights hit sum $\sum_j w_j h_{g_j}(x_0)$, by linearity of equation 6 in the columns. By Corollary 1 the conversion does not change action selection; it is required for quoting $\widehat{M}$.

4 Empirical status

Both variants were run on cube-single-play-v0, three seeds, 500k gradient steps, trained from scratch, otherwise identical to the standing recipe. Goal success is the environment's own evaluation over its five tasks at 50 episodes; the zero-shot columns apply equation 16 for Delta and equation 15 for Part.

				zero-shot reward family		
variant	seeds	goal success	diagnostic	dirac	dense	region
Delta (A)	0/1/2	.152 / .264 / .204	msq = .0042–.0100	.36–.44	—	.05–.23
Part (B)	0/2	.884 / .884	$H(p) = 3.44$ ($\log K$=5.55)	.80–.82	.70–.86	.17–.20

Table 1: Cube-single, 500k steps. Zero-shot columns are provisional (10 episodes; the protocol is 25). Seed 1 of Part is re-running after an unrelated CUDA fault. Same-environment reference points: first-passage without orthonormalization attains 0.967; visitation trained from scratch is unstable in $[0.004, 0.5]$.

Reading. Delta follows Proposition 2 exactly: msq $= \mathbb{E}[M^2]$ falls from 0.34–1.70 at 25k to 0.0042–0.0100 at 500k — two to three orders of magnitude toward zero — while the event rate $\mathbb{E}|\Delta h|$ holds flat at 0.0011 against a level $\mathbb{E}[h] = 0.0148$. The critic converges to the relocated constant. Temporal centering therefore relocates the degenerate solution and makes it legible, but does not by itself yield a usable measure.

Part passes its pre-registered gate (success ≥ 0.5 and entropy $< 0.9 \log K = 4.99$) on every seed and transfers zero-shot to point-goal reward families while degrading on extended regions. The contrast is informative because both variants share the same underlying object: what differs is the parameterization of the prediction — a normalized distribution over cells versus an unbounded scalar — and the density of supervision, not the semantics of occupancy.

Caveat. The wave contains no behavior-cloning-only control, so Part's 0.884 clears its gate but is not yet attributable to the measure term rather than the cloning term.

Relation to the multi-step note. The companion note keeps the indicator target and lengthens the operator's horizon, moving the dwell mass into the observed term; the variants here keep the one-step operator and change the predicted quantity. The two are rival explanations of the same collapse — starved bootstrap versus level-dominated regression — and are designed to be run head-to-head in identical slots.

References

L. Blier, C. Tallec, and Y. Ollivier. Learning successor states and goal-dependent values: A mathematical viewpoint. arXiv preprint arXiv:2101.07123, 2021.

Peter Dayan. Improving generalization for temporal difference learning: The successor representation. Neural Computation, 1993.

Seohong Park, Kevin Frans, Benjamin Eysenbach, and Sergey Levine. Ogbench: Benchmarking offline goal-conditioned reinforcement learning. In International Conference on Learning Representations (ICLR), 2025.

Ahmed Touati and Yann Ollivier. Learning one representation to optimize all rewards. In Advances in Neural Information Processing Systems (NeurIPS), 2021.